Controller Manual

Reflection Controller — Installation & Operation Guide

1. Purpose

The Controller provides a web-based interface to start and stop the Reflection service. This manual covers the complete setup of the Controller on a client machine, along with the steps required to start and use it afterward.

2. Prerequisites

- A pendrive containing the Controller setup zip file ("giz" package).
- Access to the target VM/server with a user account that has permission to run shell commands.
- Node.js — a bundled runtime is provided with the package, so a separate global installation is not required (see the note in Section 4).
- **Controller should be on the same vm where reflection is deployed**

3. Installation Steps

Step 1. Copy the setup package

Copy the zip file from the pendrive onto the target machine.

Step 2. Extract the package

Change into the "giz" directory and extract the archive:

```
cd <giz directory>
tar -xzvf <giz name>
```

Step 3. Move into the Controller setup directory

Change into the extracted controllerSetup directory.

```
cd controllerSetup
```

Step 4. Refresh the Monkshu symlink

Remove the old Monkshu symlink for apishell and create a new one:

```
./monkshu/rmlink apishell
./monkshu/mklink apishell
```

Step 5. Configure the apishell hostname

Edit the hostname configuration file and set the correct hostname for this deployment:

```
apishell/backend/apps/apishell/conf/hostname.json
```

Step 6. Configure SSL for apishell (if required)

If SSL is required for this deployment, edit the SSL configuration file accordingly:

```
apishell/backend/apps/apishell/conf/httpd.json
```

Step 7. Configure the Controller environment file

Edit the Controller's .env file to set the host, port, and protocol for the UI, as well as the host and protocol for apishell — matching whatever SSL configuration was set in Step 6:

```
reflection-controller/.env
```

Step 8. Rebuild npm packages

Run the install script to rebuild the required npm packages:

```
./install.sh
```

Step 9. Update the Crashguard configuration

Edit the Crashguard process configuration and set the correct paths for Monkshu, reflection-controller, and Node:

```
./crashguard/conf/process.json
```

Step 10. Allow start/restart commands for the VM user

Follow the instructions in the Reflection Controller README to grant the VM user permission to run the restart and start commands:

```
./reflection-controller/README.md
```

Step 11. Configure the Reflection service name

Edit the apishell service configuration and set the Reflection service name. This name must match the service name permitted in the command allow-list configured in Step 10:

```
./apishell/backend/apps/apishell/conf/service.json
```

Step 12. Launch the Controller

Run Crashguard to launch both the Controller's backend and frontend:

```
./crashguard/crashguard.js
```

Note: controllerSetup/node-runtime/bin/node is Node v24.0 and can be used if you do not want to install Node 24 globally on the machine (/usr/bin/node). The install.sh script uses node-runtime/bin/node for the npm rebuild by default, and the default Crashguard configuration also points to node-runtime for Node by default.

4. Using the Controller

Once Crashguard has launched the Controller's backend and frontend (Step 12), open the Controller UI in a browser using the host, port, and protocol configured in the .env file (Step 7).

From the UI, the Reflection service can be started or stopped using the corresponding controls. These actions are carried out through apishell, using the service name configured in Step 11.

5. Verification

- Confirm the Controller UI loads at the configured host and port.
- Confirm the Start/Stop actions correctly start and stop the Reflection service on the machine.
- If an action fails, re-check the service name in service.json (Step 11) and the command allow-list from the README (Step 10).
- If the UI does not load, re-check the .env values (Step 7) and the hostname/SSL configuration (Steps 5–6).